McMurdo Station, Ross Island

BUILDING INFORMATION SHEET

Water Plant

Location ID: B198

Functions as the primary water desalination plant and fresh water holding facility. Contains three 40,000 gallon per day reverse osmosis processors, three 50,000 gallon fresh water storage tanks, and one 18,000 gallon seawater surge tank. The liquid nitrogen plant is also located in Building 198.

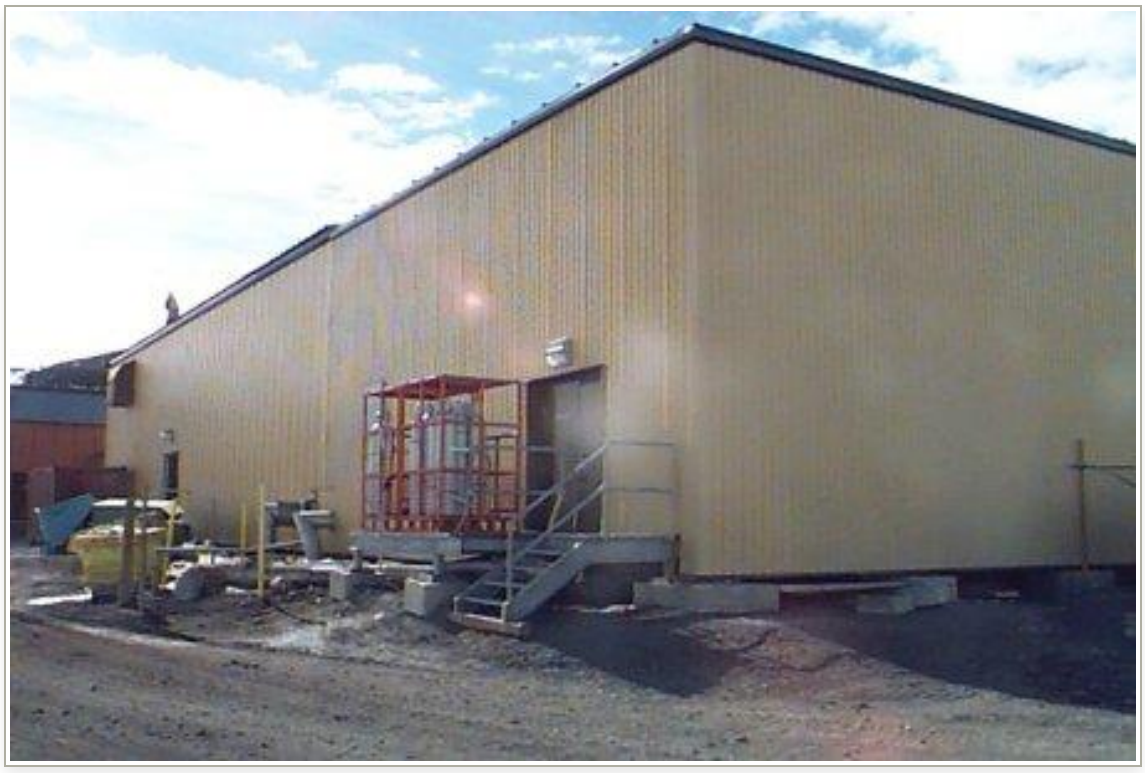


Photo 1: *Water Plant*

General Structure Information



Automatic External Defibrillator (AED) Locations

This building contains one or more AED devices that may be used in the event of life-threatening cardiac arrhythmia.

- **Western wall of generator room, adjacent to the CO2 suppression cylinders**

Usage	Water Plant
Occupancy Group	B-4 Pumping Plant
Maximum Occupancy	24
Construction Type	Type II, non-combustible; steel-framed structure with foam insulated metal siding and roofing, steel-framed precast concrete slab floor, no interior partitions. A 60' x 40' addition is of similiar construction.
Year Constructed	1984-87
Dimensions	60' x 80' and a 60' x 40' addition

Floor Area

Floor	Area
Floor 1	7,200 sq ft
Floor 2	1,000 sq ft
Total	8,200 sq ft

HVAC System

Heating	Waste heat back-up boiler located in the Water Plant is a Superior steam boiler converted to hot water. The Water Plant back-up furnace is a Jackson and Church 500,000 Btu/hr unit.
Air Handling Units	None
Fuel Tank	Two 20,500 gallon fuel tanks located behind Buildings 196 and 198

Water and Sewer

Water Service	Not provided on water distribution system
Sewer Service	Not connected
Connections	Sinks: 1 Toilets: 0 Urinals: 0 Emergency Eyewash: 1 Water Cooler: 1 One line with one instrument
Water Heater	6 gallon electric water heater

Electrical Service

Service Feeder	Feeder F
Electrical Service	208/120V, 3PH, 1,200A main breaker
Phone and LAN	One line with one instrument

Fire System

Fire Panel Type	King-Fisher KFRTI-20 industrial radio and control system (IRACS) panel, pull stations, alarm horns
Heat Detectors	Located throughout building
Suppression Type	Automatic dry-chemical fire extinguisher located over the furnace and boiler.

Floorplans

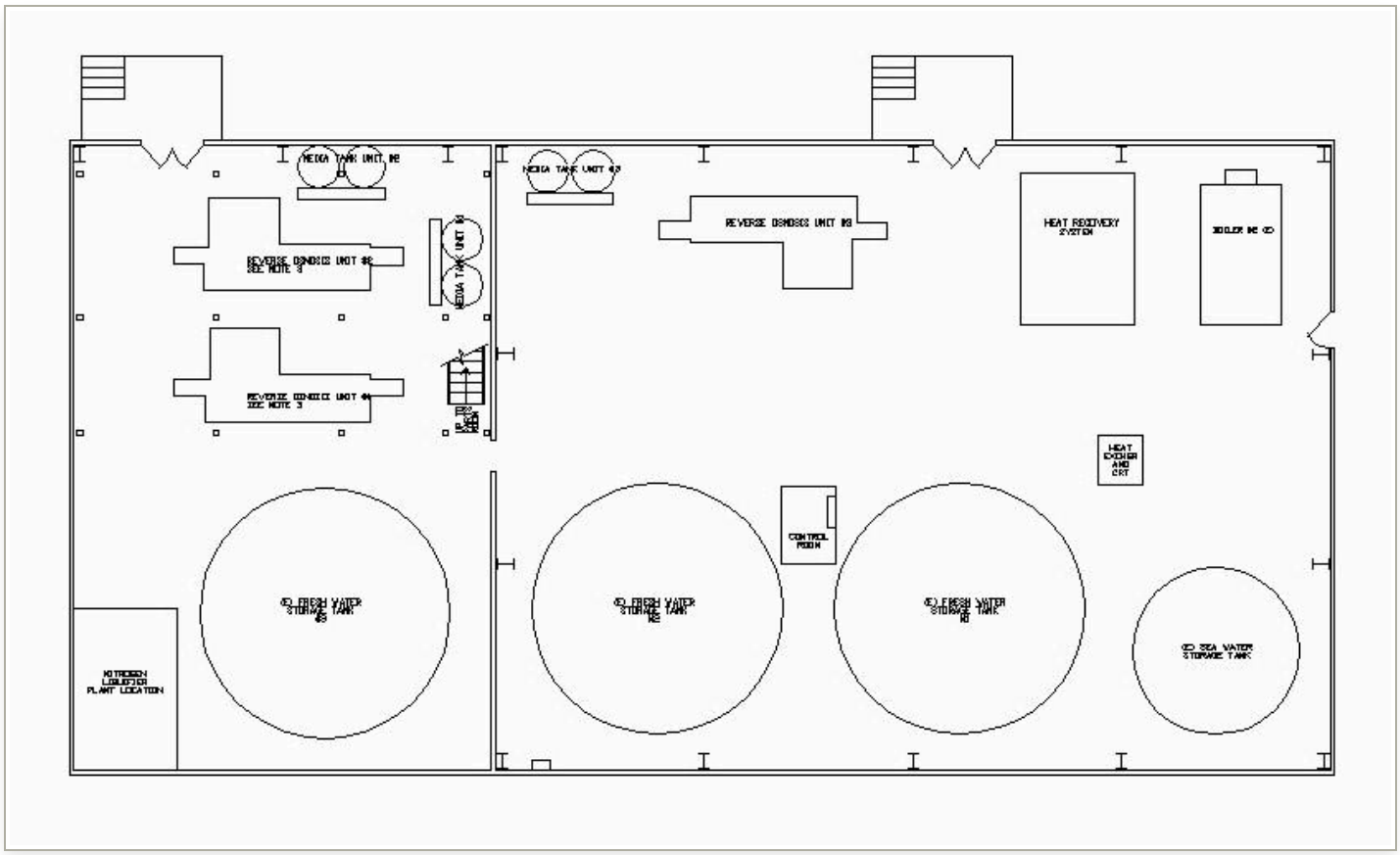


Figure 1: *Water Plant*